Algorithms

Lecture #44

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Reductions

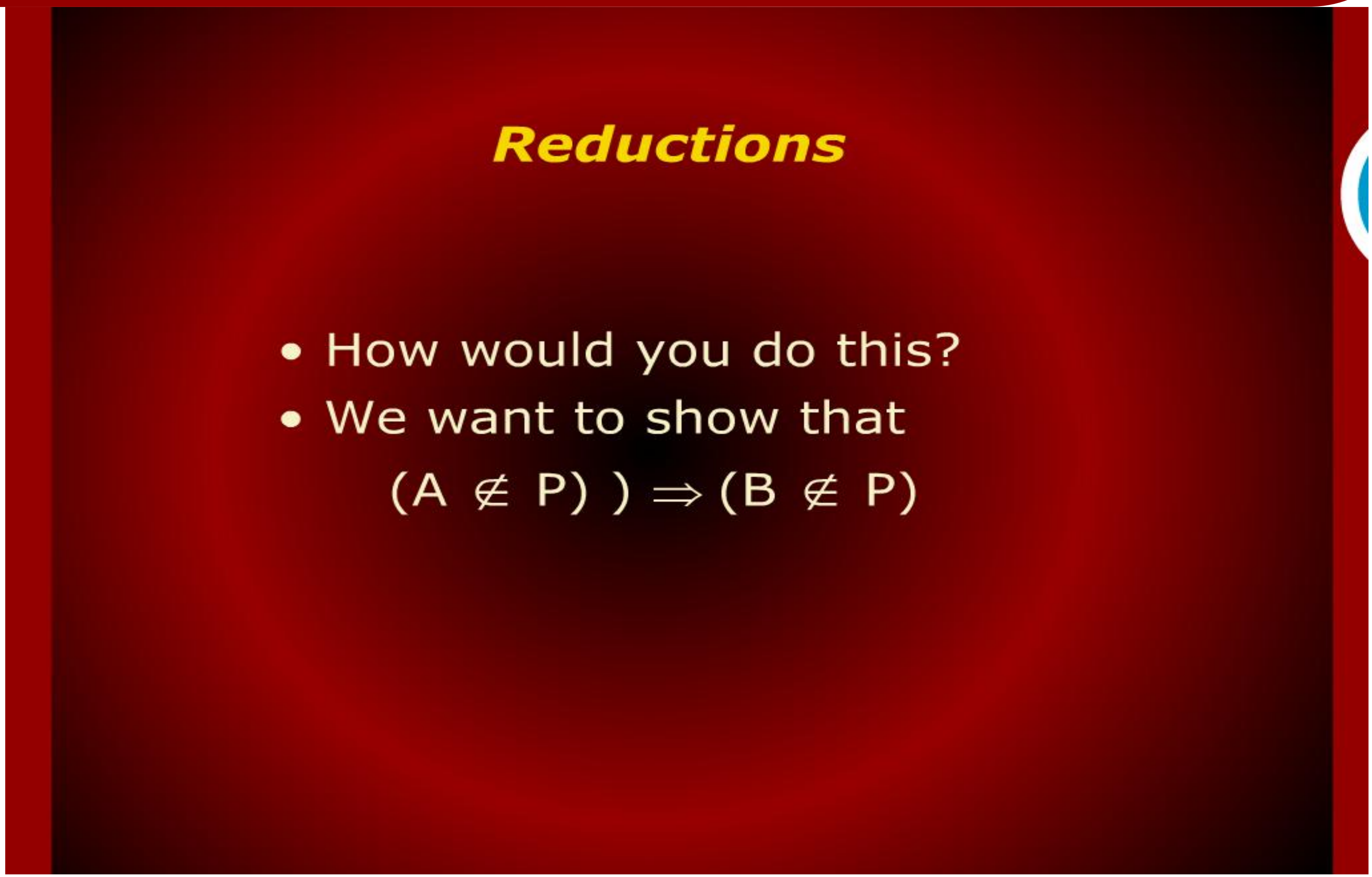
Reductions

Reductions

Consider the question:

- Suppose there are two problems, A and B.
- You know (or you strongly believe at least) that it is impossible to solve problem A in polynomial time.
- You want to prove that B cannot be solved in polynomial time.

Reductions



Reductions

Reductions

Consider an example to illustrate reduction:

The following problem is well-known to be NPC:

3-color: Given a graph G, can each of its vertices can be labelled with one of 3 different colors such that two adjacent vertices have the same label (color).

Reductions

Reductions

- Coloring arises in various partitioning problems where there is a constraint that two objects cannot be assigned to the same set of partitions.
- The term “coloring” comes from the original application which was in map drawing.
- Two countries that share a common border should be colored with different colors.

Reductions

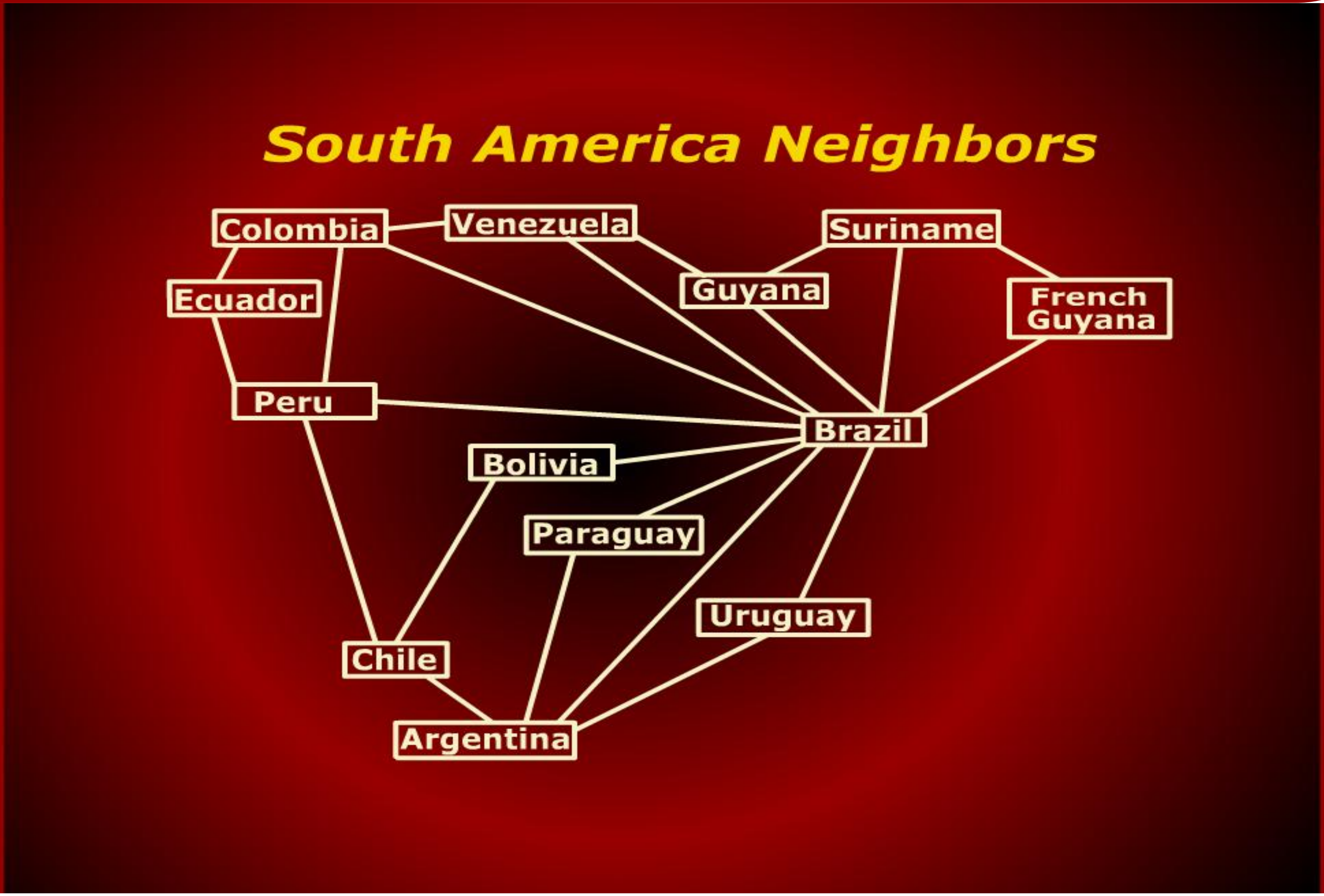
Reductions

- It is well known that planar graphs can be colored (maps) with four colors.
- There exists a polynomial time algorithm for this.
- But determining whether this can be done with 3 colors is hard and there is no polynomial time algorithm for it.

Reductions

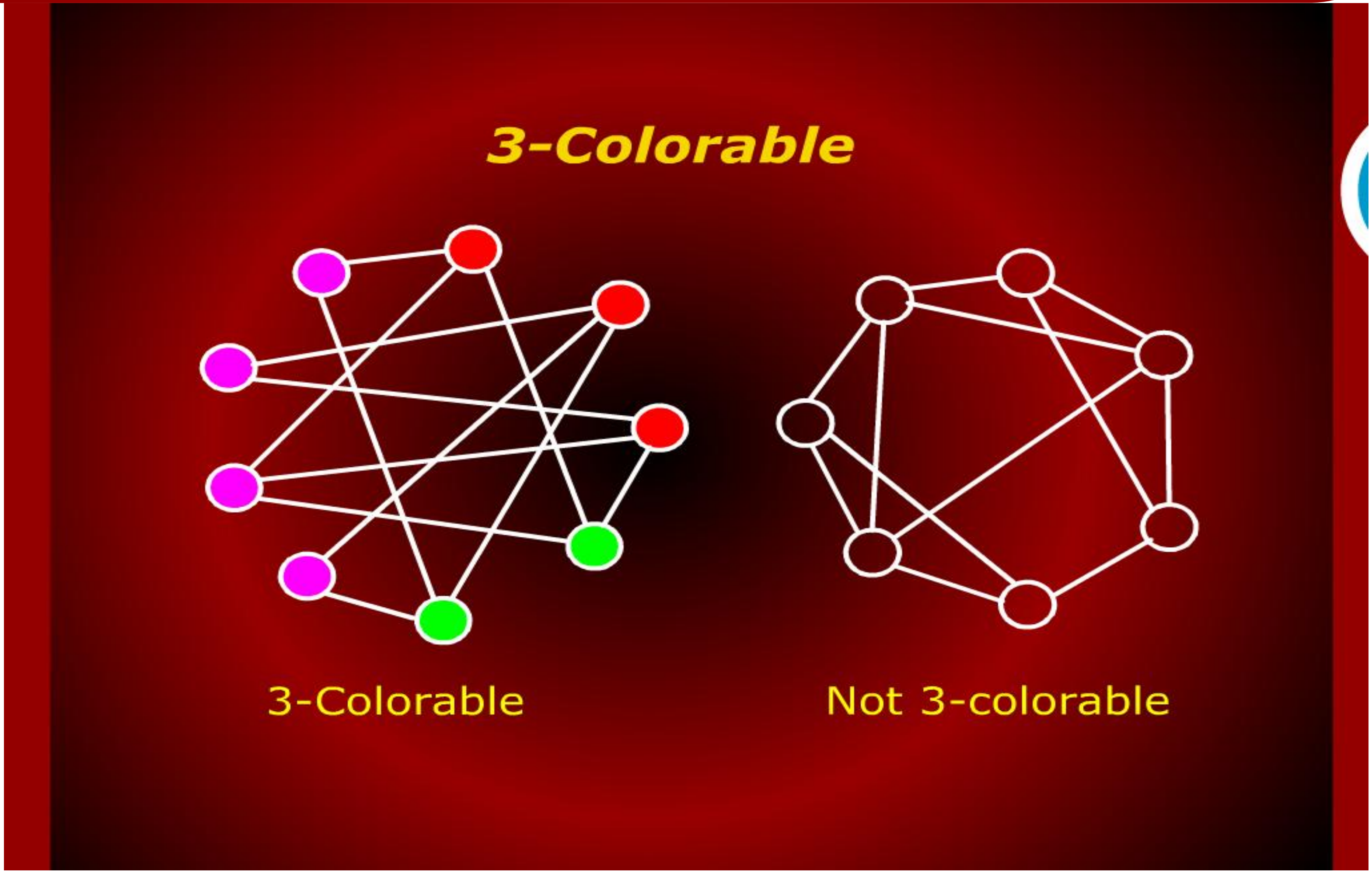

Reductions
VENEZUELA
COLOMBIA
PERU
BRAZIL
BOLIVIA
ARGENTINA
793-1

South America Neighbors



USA Map

3-Colorable



Graph Coloring: Fish Tank

Graph Coloring: Fish Tank

- A tropical fish hobbyist had six different types of fish designated by A, B, C, D, E, and F, respectively.
- Because of predator-prey relationships, water conditions and size, some fish can be kept in the same tank.

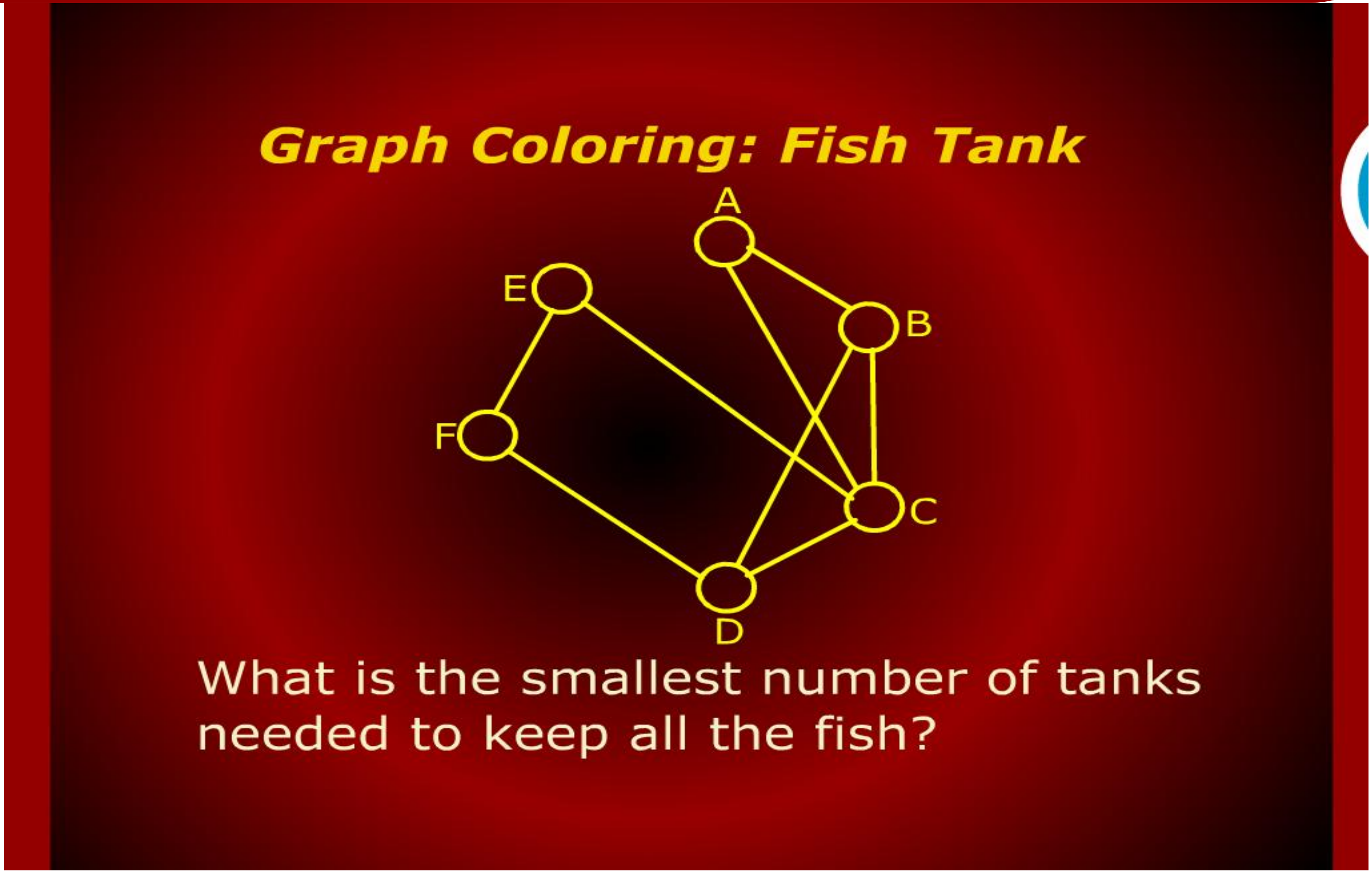
Graph Coloring: Fish Tank

Graph Coloring: Fish Tank

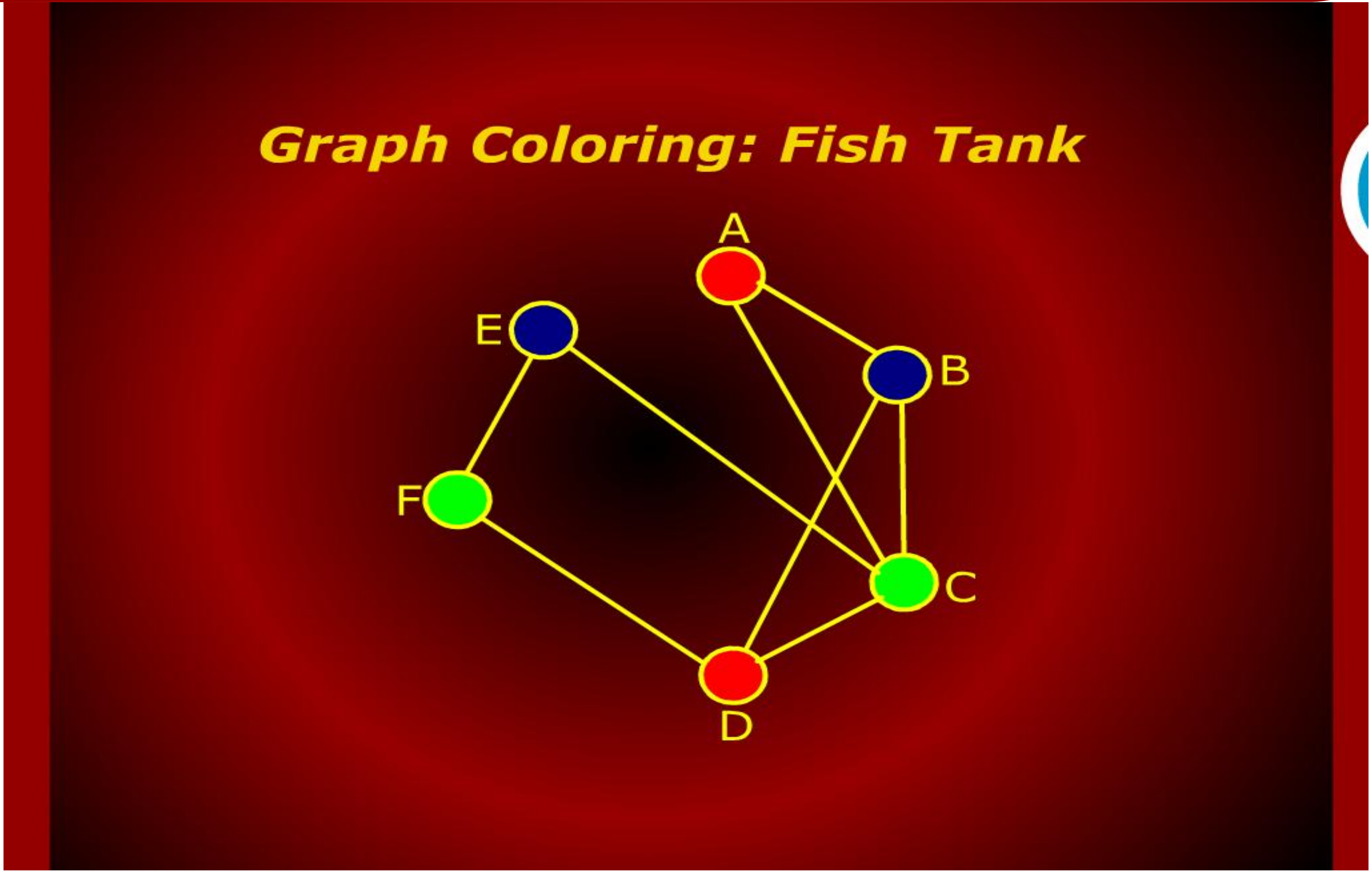
- The following table shows which fish cannot be together:

Type	Cannot be with
A	B, C
B	A ,C, E
C	A, B, D, E
D	C, F
E	B, C, F
F	D, E

Graph Coloring: Fish Tank



Graph Coloring: Fish Tank



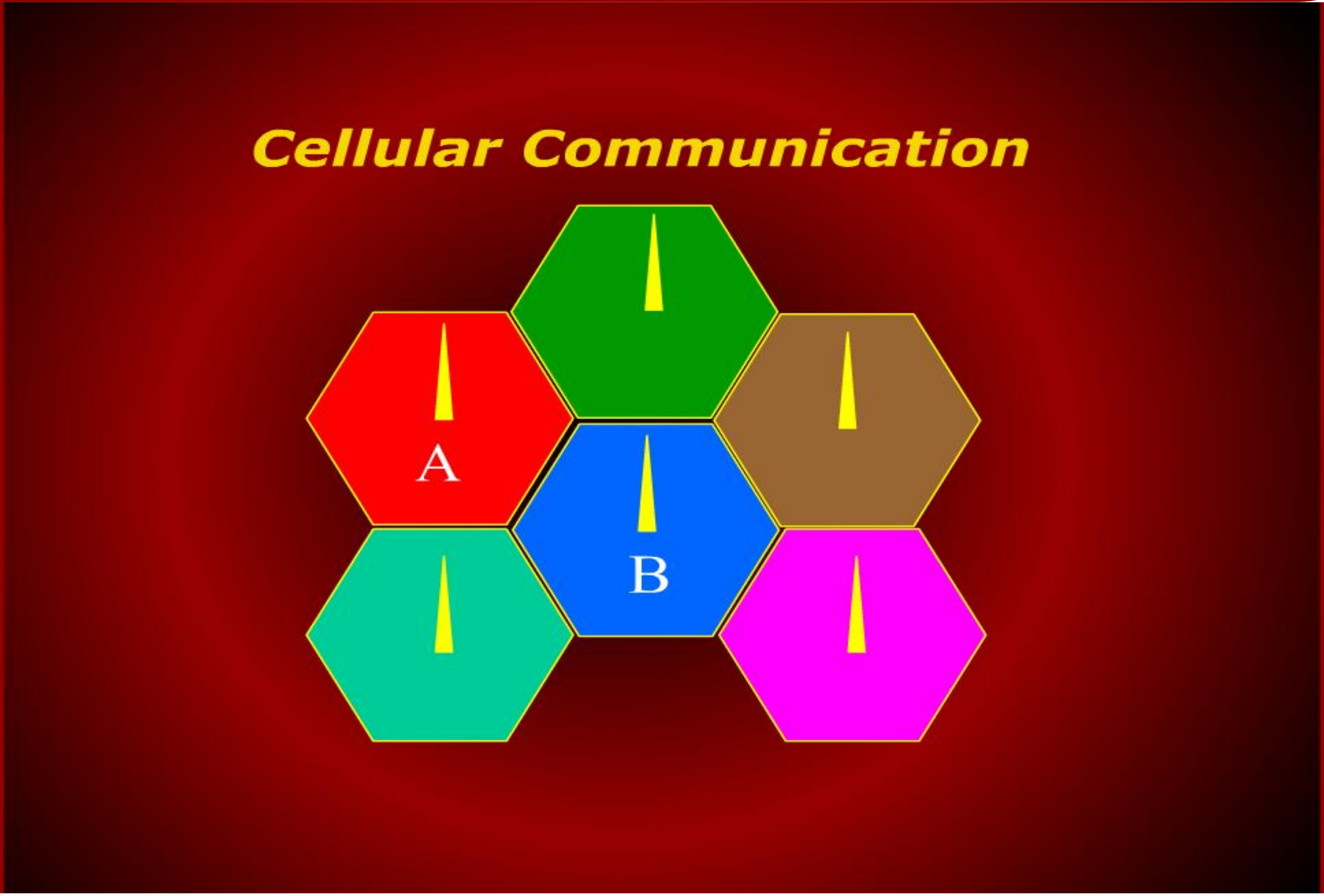
Graph Coloring: Fish Tank

Graph Coloring: Fish Tank

The fewest number of tanks the tropical fish owner will need is three.

Tank 1	Tank 2	Tank 3
A, D	F, C	B, E

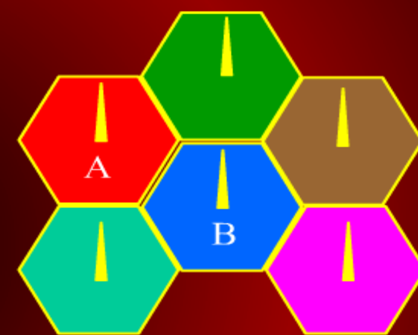
Cellular Communication



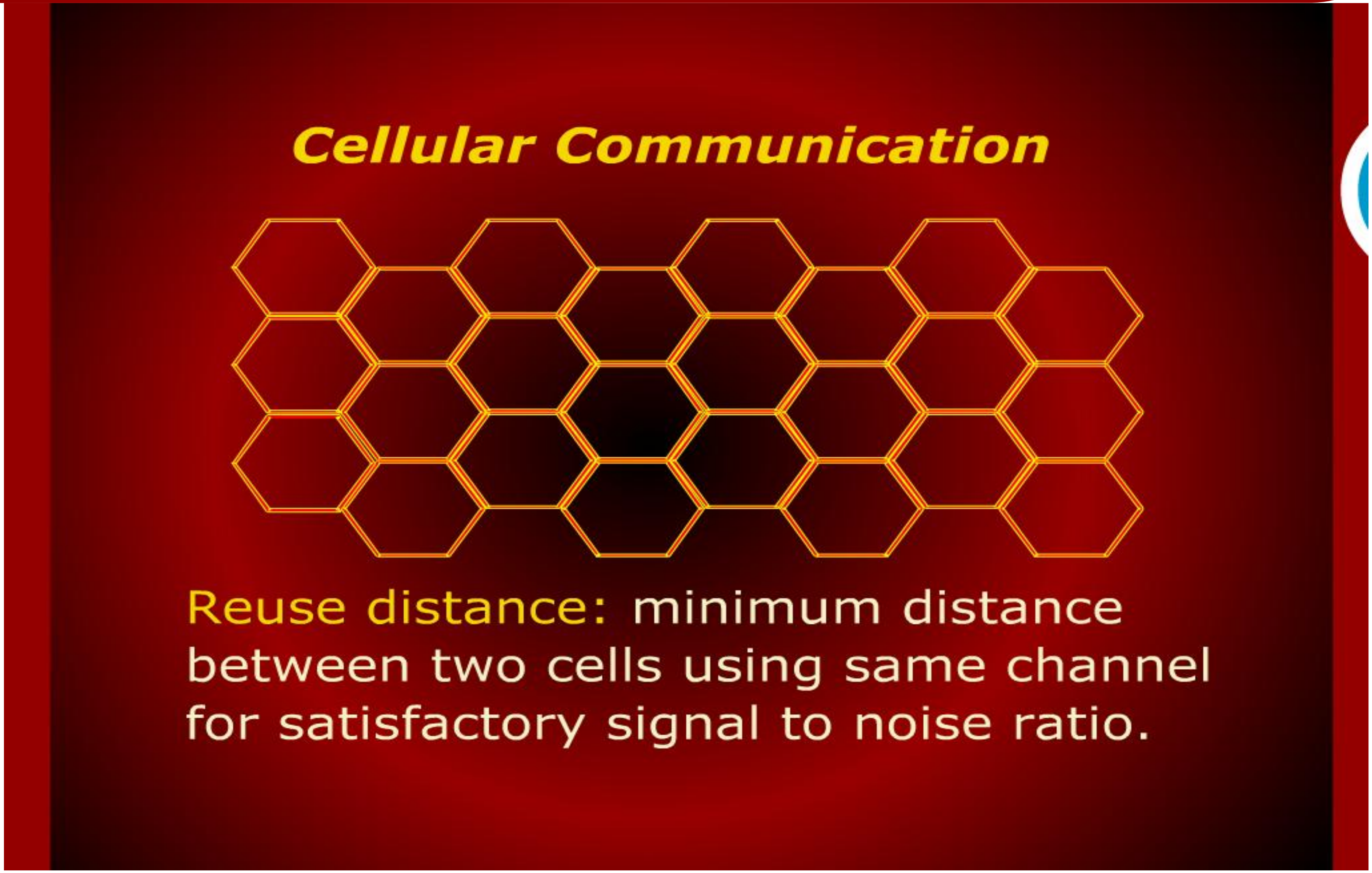
Cellular Communication

Cellular Communication

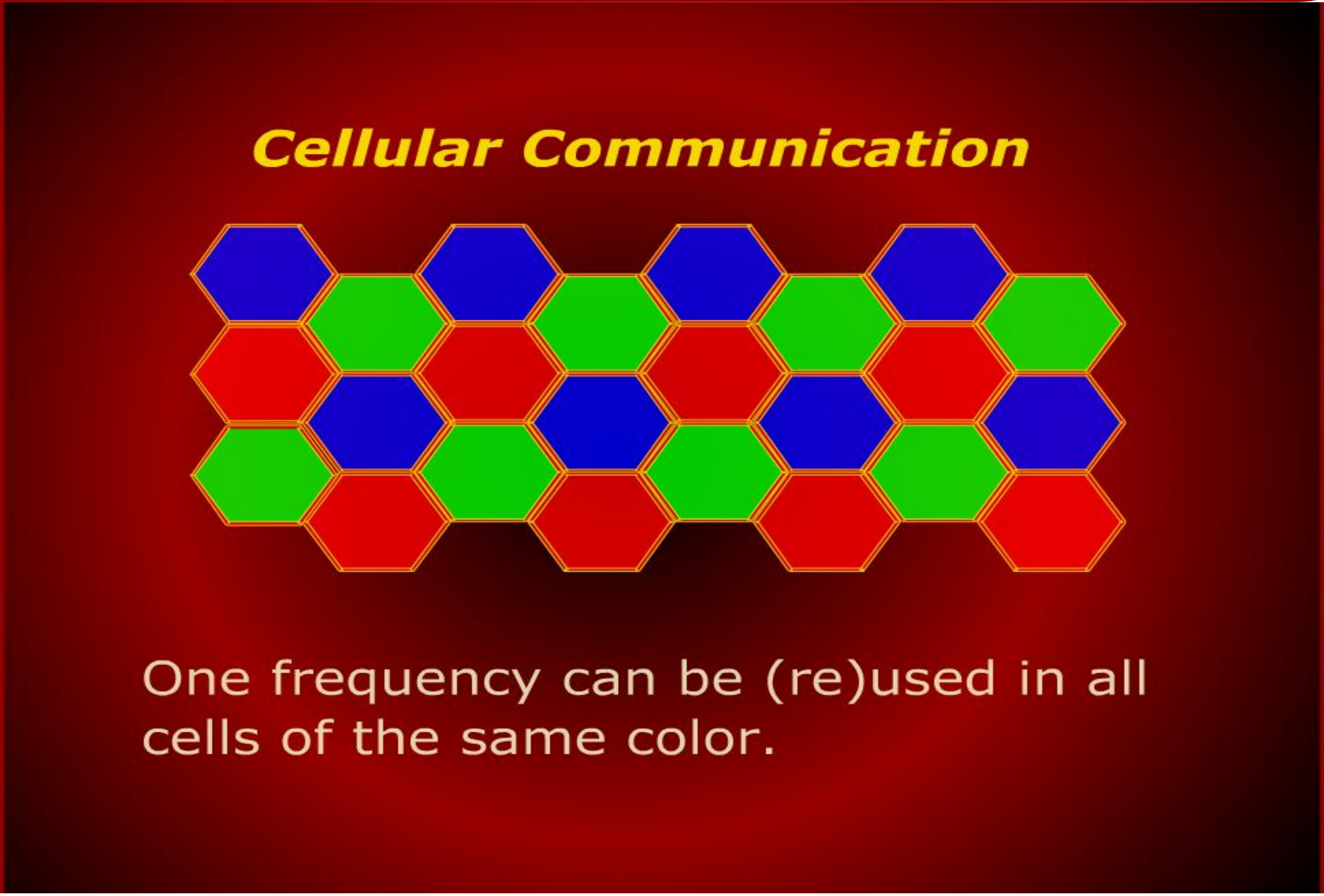
- One low power transmitter per cell
- Frequency reuse: limited spectrum
- Handoff: moving to other cells
- Co-channel interference: transmission on same frequency



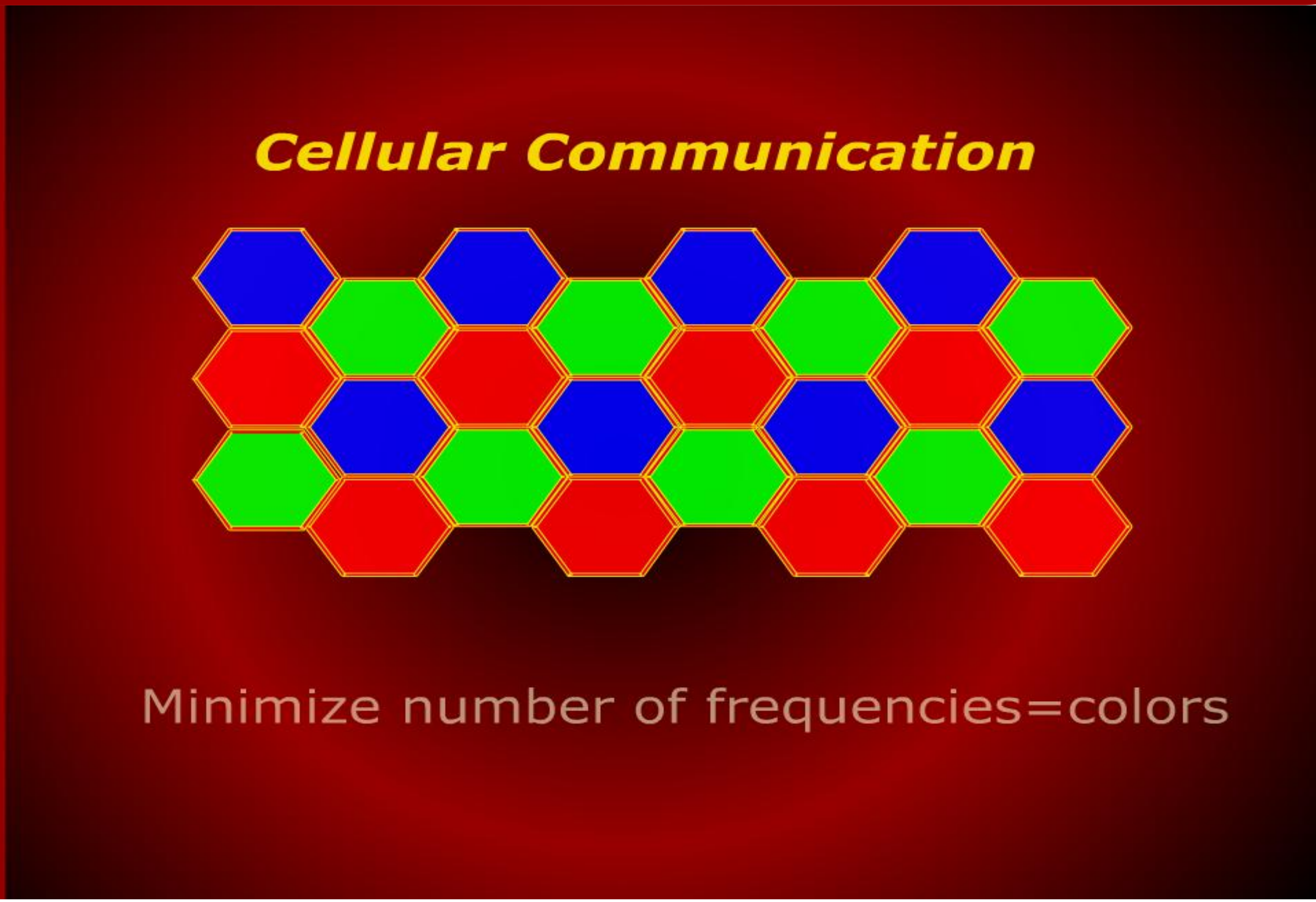
Cellular Communication



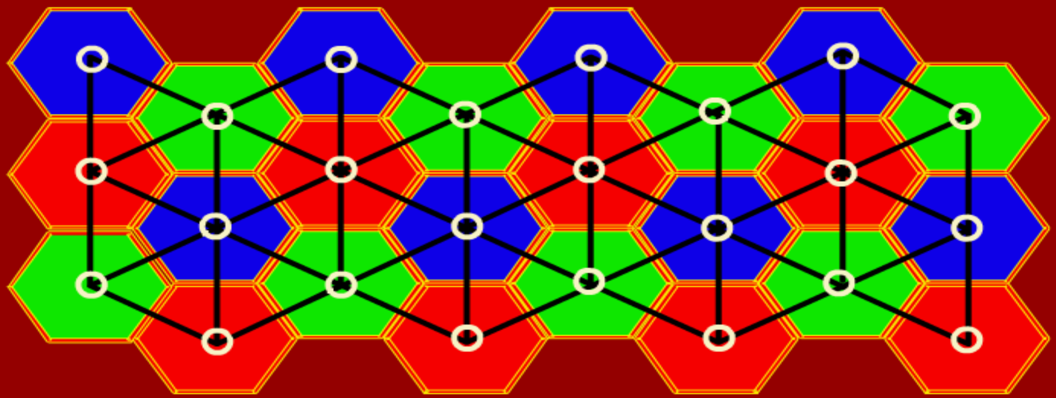
Cellular Communication



Cellular Communication

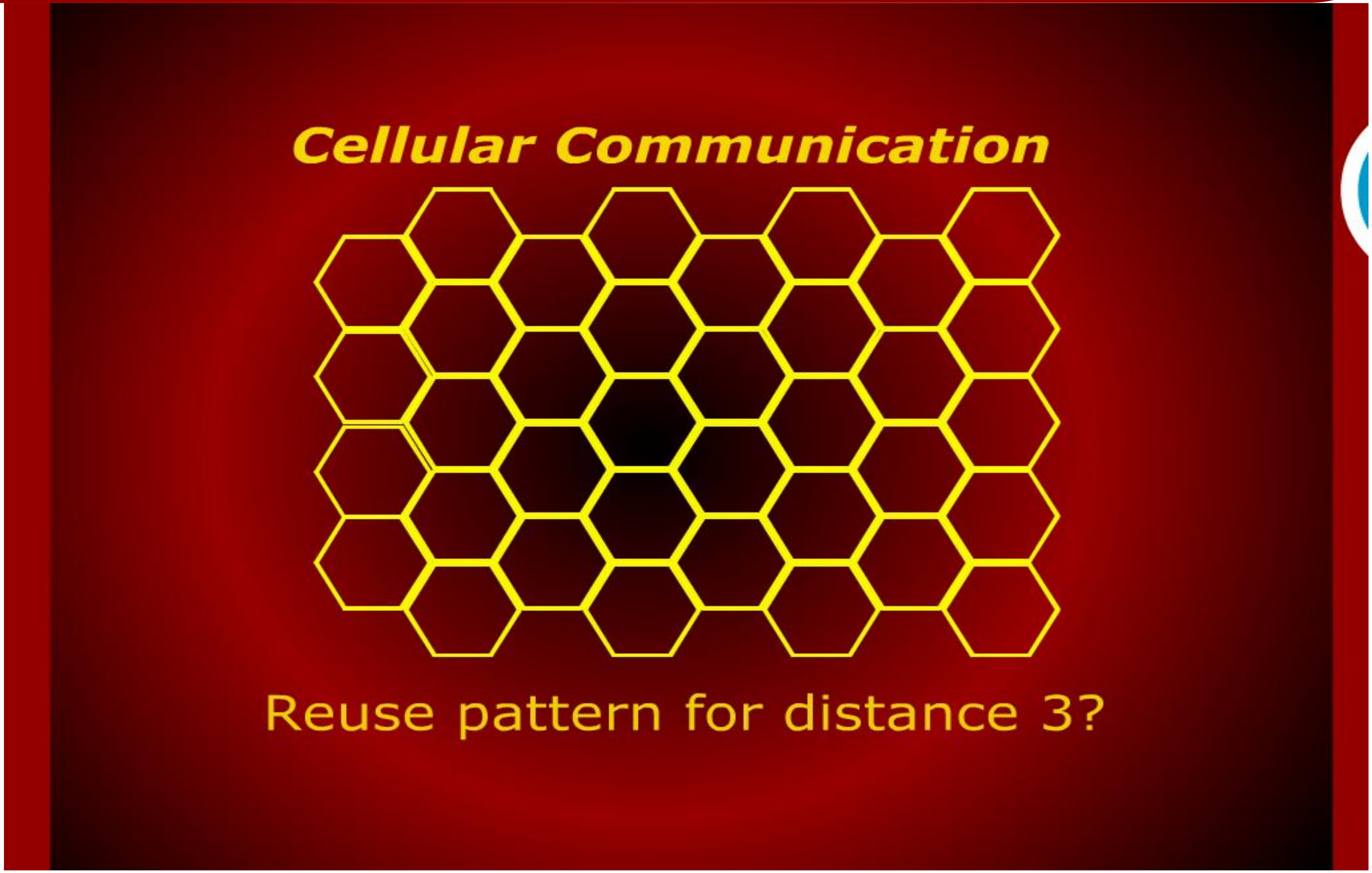


Cellular Communication

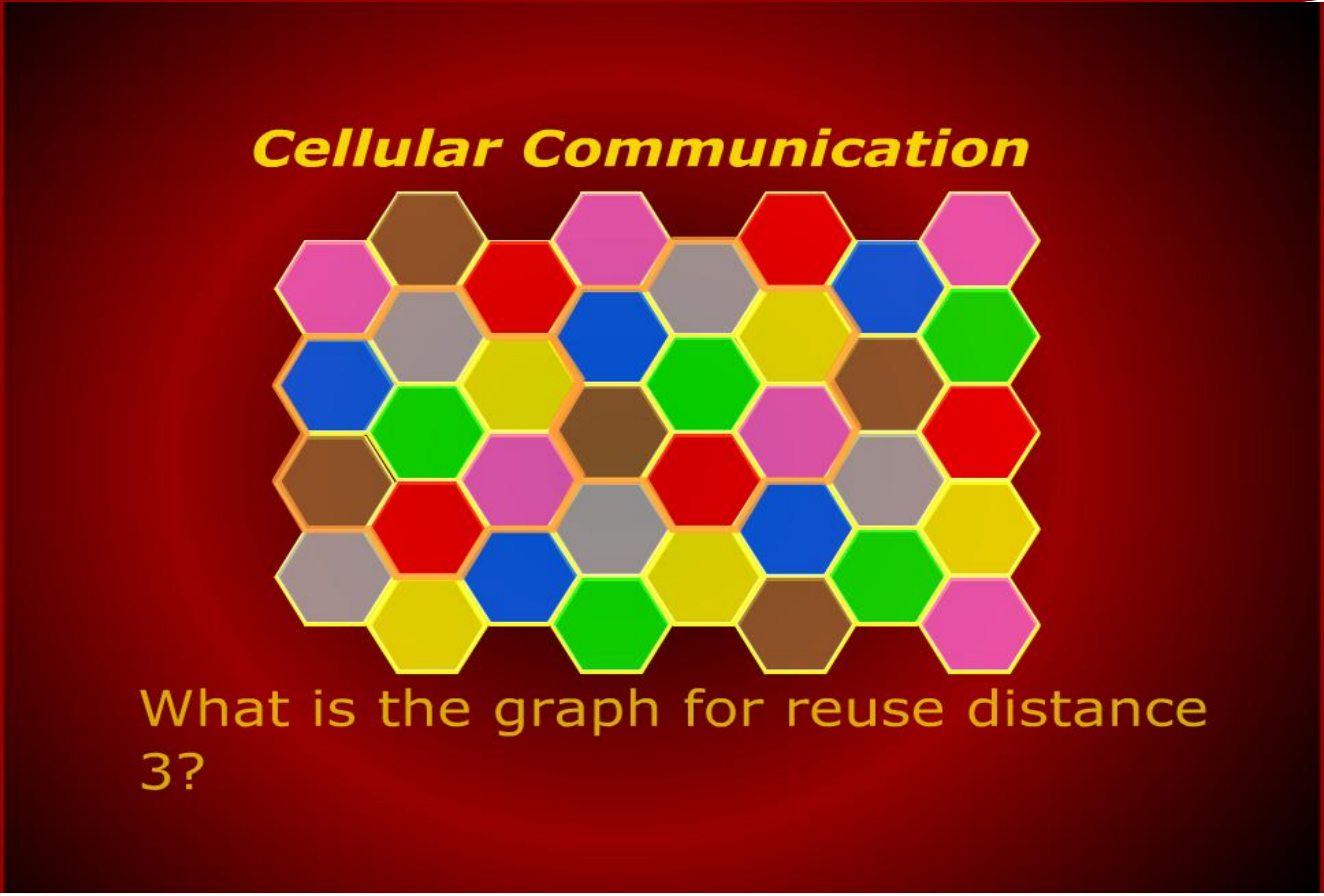


What is the graph for reuse distance 2?

Cellular Communication



Cellular Communication



Reductions

Reductions

- The 3-color (3Col) problem will the play the role of A, which we strongly suspect to not be solvable in polynomial time.
- For our problem B, consider the following problem: